

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

State the principle of conservation of momentum and apply it to a head-on elastic collision.

[6 marks]

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

2

A resistor network has two $6\ \Omega$ resistors in parallel with a $3\ \Omega$ resistor in series. Find the total resistance.

[5 marks]

3

Sketch and explain the shape of a displacement-time graph for a wave, and state what the axes tell you.

[5 marks]

4

Calculate the energy released when a body of mass 2.0 kg falls through 15 m, stating any assumption.

[4 marks]

$$E = mgh$$

5

Describe an experiment to measure the acceleration due to gravity, identifying the main source of error.

[6 marks]

6

A 60 kg skater pushes off a 40 kg skater and moves at 2 m/s. Find the other skater's speed.

[5 marks]

7

Explain the difference between distance and displacement using a closed-loop journey.

[4 marks]

8

Calculate the current through a $12\ \Omega$ resistor connected across a 9 V supply, and the power dissipated.

[5 marks]

$$V = IR$$
